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The way towards European electricity intraday auctions – Status quo and future developments

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ABSTRACT

This paper sheds light on the status quo of currently implemented electricity intraday auctions in Europe and offers an outlook for future developments. First, we compare the two market mechanisms “continuous trading” and “auction” and identify advantages and disadvantages. Then, we investigate the currently existing six intraday auctions in Europe. We compare crucial auction characteristics such as the number of auctions, tradable market period(s), gate opening time and gate closure time, and find a wide variety in auction designs. By examining relevant European regulation and recent regulatory decisions, we illustrate that future European intraday auctions can either be implemented as cross-border auctions or complementary regional auctions. We find that complementary regional auctions of the borders Portugal-Spain, CCR Greece-Italy and CCR Italy-North are already approved.

1. Introduction

In most European countries the current short-term electricity market sequence consists of day-ahead and intraday markets, often referred to as “spot markets” (e.g. Zweifel et al., 2017; EPEX SPOT SE, 2020a). Their market designs differ in several aspects. For instance, day-ahead markets are coupled across most of Europe since several years (“Single Day-ahead Coupling”; e.g. ENTSO-E, 2020), while intraday markets only recently underwent changes from a national to a European scope. Furthermore, there is a difference with regard to price formation: day-ahead markets are commonly organized as auctions, i.e. aggregation of demand and supply, whereas intraday markets are usually implemented as continuous trading.[1] The latter enables market parties to balance supply and demand changes in the short-term to reduce exposure to imbalance penalty (e.g. Chaves-Ávila and Fernandes, 2015; Soysal et al., 2017; Ehrenmann et al., 2019). These changes occur because contracted quantities in the day-ahead timeframe are based on forecasts, which are prone to error: they depend on different parameters, most notably on the intermittent production from volatile renewable energy sources (henceforth referred to as “VRES”) such as wind and solar power plants. Closer to real-time, forecasts are more accurate and market parties can adjust positions directly on a bilateral basis by continuous trading (Zweifel et al., 2017).

In the light of the growing production share by VRES across Europe, this both national and bilateral manner of intraday trading may no longer be sufficient. Lanfranconi, Lanza and Rossi (2019) argue that “economic arguments suggest that the current intraday market design at European level (…) would benefit from improvements”. Ehrenmann et al. (2019) state that the very purpose of intraday markets needs to be re-considered: they will no longer be only (national) adjustment markets because the moment of exchange is moving closer to real-time. Consequently, intraday trading should be revised and developed in such a way that the integration of VRES in the system is facilitated by all available means of market design (e.g. ACER/CEER, 2018). We pick up on these arguments and discuss two options in more detail: firstly, the coupling of existing intraday continuous markets in Europe, which enables market parties to balance their positions also across borders. Secondly, and in the focus of this paper, the implementation of auctions– on a national and European level –in the intraday time frame, which provides an alternative to bilateral, continuous trading.

The merger of continuous intraday markets to a coupled cross-border European market was established in June 2018. It is known as the cross-border intraday (henceforth referred to as “XBID”) system (ENTSO-E,

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[1] Lanfranconi et al. (2019) argue that the “choice of continuous trading as the European target model for intraday (…) comes from historical reasons” because it was the applied mechanism in Northern and Central Europe at the time of market integration. Note that further differences are discussed in Section 3.

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Abbreviations

ACER	Agency for the Cooperation of Energy Regulators
CACM	Regulation Guideline on Capacity Allocation and Congestion Management
CCR	Capacity Calculation Region
CRIDA	Complementary Regional Intraday Auction
D	Day of delivery
D-1	Day before delivery (day-ahead)
IDA	Intraday Auction
NRA	National Regulatory Authority
SIDC	Single Intraday Coupling
TSO	Transmission System Operator
XBID	Cross-border Intraday Trading
VRES	Volatile Renewable Energy Sources

2018; Amprion, 2020). The XBID system initially involved 14 countries and represents the IT implementation of the "single intraday coupling" (henceforth referred to as "SIDC") for continuous trading, which was recently promoted by European guidelines and network codes (European Commission, 2015, 2019).[2] The XBID system unites the continuous intraday markets and complements the already coupled day-ahead markets. In November 2019, the XBID system was enlarged by seven countries: now the total number of coupled countries is 21 (SIDC, 2019; HUPX, 2019).[3] This recent enlargement illustrates that continuous intraday trading was until now in focus of European policy makers.

The second option, the introduction of auctions in the intraday timeframe, has caught far less attention of policy makers until now (e.g. Bellenbaum et al., 2014; Lanfranconi et al., 2019). Only some European countries apply intraday auctions (henceforth referred to as "IDAs") instead of or as a complement to continuous trading such as, for instance, Italy, Spain and Portugal do since the early 2000s (e.g. GME, 2003; Furió, 2011; Chaves-Ávila and Fernandes, 2015). Nonetheless, the European guideline on capacity allocation and congestion management (henceforth referred to as "CACM Regulation") and recent decisions by the Agency for the Cooperation of Energy Regulators (henceforth referred to as "ACER") present general framework conditions for future European IDAs (European Commission, 2015; ACER, 2017, 2019a). The overarching objective is to complement the SIDC for continuous trading (ACER refers to this as "auction SIDC").

The aim of this paper is to present the status quo of IDAs in Europe, to address recent regulatory stipulations and to give an outlook for future developments. Our approach can be divided in three parts. First, in Section 3, the theoretical background is presented: we discuss the differences between the two market mechanisms regarding liquidity, market power resilience, information efficiency, static allocation efficiency and the efficient usage of cross-zonal capacity, and advantages and disadvantages are concluded. Second, we carry out an empirical analysis of existing IDAs in Europe in Section 4. We consider relevant characteristics to compare the auctions, such as the number of auctions, tradable market period(s), gate opening time and gate closure time, and show that there exists a wide variety in currently implemented auction designs. Third, we illustrate and discuss the regulatory framework conditions for future European IDAs in Section 5. We present that future European IDAs can either be implemented as cross-border IDAs or as complementary regional IDAs. Furthermore, we refer to recent applications of these stipulations in Southern European countries: the complementary regional IDAs at the borders Portugal-Spain, Greece-Italy and Northern Italy are already approved.

The remainder of this paper is structured as follows: Section 2 presents related literature and background information, and Section 3 discusses the theoretical background. In Section 4, we present our analysis of the currently existing European IDAs, and Section 5 illustrates the regulatory conditions for future European IDAs. Section 6 concludes and draws policy implications.

2. Background and literature review

This literature review focuses on IDAs and presents background information regarding the current state of discussion.[4] Despite the only occasional implementation of IDAs in Europe, there are many scientific papers and documents around this topic.[5]

Regarding Southern European IDAs, being operational already for a long time, Furió (2011) presents an econometrical analysis of the Spanish IDA from 2000 to 2010, focusing on prices and the evolution of the registered electricity traded. It is shown that a growing interest in intraday trading exists, particularly in the last-time-negotiated hours. Chaves-Ávila and Fernandes (2015) analyze the participation of renewable generators in the Spanish intraday market. They explore different market rules in the Spanish market that incentive market parties to participate and find that the market design has contributed to renewable generation balancing. Lanfranconi, Lanza and Rossi (2019) present a recent analysis of IDAs across bidding zones in Italy and between Italy and Slovenia. By providing an empirical analysis, they suggest to re-consider the role of IDAs because they allow for more efficient use of cross-zonal capacity.

With respect to the German IDAs, being operational since the end of 2014, Weber (2010) investigates the major European power markets France, Germany, Scandinavia, Spain and UK with a particular focus on liquidity. One proposal for improving liquidity in the German market is the introduction of IDAs. Neuhoff et al. (2016a) investigate the question whether adding auctions to the continuous intraday trading is improving market performance. They assess this by referring to the experience with the implementation of the German IDA, with a focus on trading volumes, prices and market depth. Ocker and Ehrhart (2017) link the declining demand for balancing power in Germany to the introduction of the German IDA. Braun and Brunner (2018) study the quarter-hourly IDA in the German market and find a distinctive zigzag price formation, which they explain by two factors: "first, the solar residual that combines the trading of solar power ramps around midday as well as the gradients of consumption and thermal power plant ramps throughout the course of the day, and second, a characteristic two stage market design with higher liquidity for the hourly than for the quarter-hourly auction."

Regarding theoretical analyses and market design questions, Henriot (2012) examines the benefits for wind power producers to trade in intraday markets. It is argued that in some cases it may be inefficient to implement IDAs: market participants adjust their strategies to the inflexible points of trading, which may lead to additional costs and lost trading opportunities. Bellenbaum et al. (2014) discuss various options to use and price intraday cross-zonal capacity in the context of the requirements of the European target model and the network codes.

[2] The 14 countries are Austria, Belgium, Denmark, Estonia, Finland, France, Germany, Latvia, Lithonia, Norway, Portugal, Spain, Sweden and the Netherlands.

[3] The so-called "second wave go-live" in XBID includes Bulgaria, Croatia, Czech Republic, Hungary, Poland, Romania and Slovenia. A "third wave go-live" is planned at the end of 2020 with Italy and Greece (HUPX, 2019).

[4] Literature on continuous trading can be found in Diamantopoulous et al. (2012); Hagemann and Weber (2013); Skajaa et al. (2015); Just and Weber (2015); Scharff and Amelin (2016); Kiesel and Paraschiv (2017); Soysal et al. (2017); Martin and Otterson (2018); Märkle-Huß et al. (2018); Maciejowska et al. (2019) or Shinde and Amelin (2019).

[5] Further literature can be found in Bellenbaum et al. (2014) or Lanfranconi et al. (2019).

Neuhoff et al. (2015) discuss the use of IDAs within and between European member states. They conclude that IDAs could present "an important step to enhance efficiency thus reducing cost of operation of European power systems, to accommodate increasing shares of wind and solar energy, and to enhance security of system operation." In an extensive study by IRENA (2017), continuous trading is contrasted with IDAs in various aspects and a possible co-existent market design is discussed. Ehrenmann et al. (2019) argue that European IDAs will become indispensable in the light of the rising share of electricity production by VRES. They further argue that auction may foster the participation of new and smaller market participants. It is recommended to introduce several cross-zonal IDAs, such as an opening auction, an evening auction and additional auctions in between.

3. Theoretical background

Any market-based solution shall reach the overarching political goal of welfare maximization (e.g. European Commission, 2015, 2016, 2017, 2019). In literature, several criteria exist to evaluate this aim.[6] We focus on the most prominent ones: liquidity, market power resilience, information efficiency, static allocation efficiency and efficient usage of cross-zonal capacity. We conclude this section with an overview of advantages and disadvantages.

3.1. Continuous trading and auction mechanism

In both mechanisms market parties submit ask and offer bids to the auction office, which are published in a central order book (Zweifel et al., 2017). The principle of continuous trading builds upon an instantaneous matching of suitable bids until gate closure, while the auction principle aggregates, sorts and clears the entire market at once after gate closure (Henriot, 2012).[7] For auctions, the price rule can either be uniform pricing (pay-as-cleared), i.e. all awarded market parties pay/receive the same price, or pay-as-bid pricing, i.e. all awarded parties pay/receive their individual bids. For continuous trading it is limited to pay-as-bid pricing (Neuhoff et al., 2015). Note that there are both national (Braun, 2018) and international continuous intraday markets (e.g. the XBID system presented in the Introduction).

3.2. Liquidity

ACER/CEER (2018) define market liquidity as "a key indicator of a well-functioning electricity market" that can be defined as "the feature of the electricity market whereby a large number of market participants are able to sell/buy products quickly, without significantly affecting the product's price and without incurring significant transaction costs". Bellenbaum et al. (2014) state that a high liquidity is advantageous for the overall efficiency because it means a high number of market participants and high trading volumes. From a general viewpoint, IRENA (2017) argues that auctions present the advantage of higher liquidity because they concentrate transactions since the last trading session.[8] Lanfranconi et al. (2019) point to an analysis of intraday traded volumes per product and per bidding zone for the largest European markets in 2017, which supports this reasoning: four of six bidding zone borders with the highest intraday trading volumes implemented IDAs.

3.3. Market power resilience

Mas-Colell, Whinston and Green (1995) define market power as "(…) the ability to alter profitably prices away from competitive levels". This means that, in contrary to competitive markets, not all consumers and producers act as price takers, i.e. the demand and supply are not infinitely elastic at going market prices. Bellenbaum et al. (2014) argue that continuous trading is of advantage for larger firms because they have a "better return on information costs and thus creates barriers to entry". They conclude that all investigated continuous trading options are more prone to market power than auction solutions. IRENA (2017) argues that this advantage for auctions only holds as long as sufficient liquidity can be guaranteed.

3.4. Information efficiency

Bellenbaum et al. (2014) describe information efficiency as follows: "[the] arrival of new information is instantaneously translated into market adjustments". This means that, for instance, a change in wind forecast is immediately reflected into price signals. This is enabled by continuous trading because any new information can be processed directly into a new trade (IRENA, 2017). On the contrary, auctions are conducted only at certain points in time: information can only be translated into price signals with a delay that is given by the timespan between the arrival of the new information and the time of auction clearing.[9] Lanfranconi et al. (2019) summarize it as follows: "(…) in order to grant efficient allocation (…) auctions are in principle needed each time a change in scarcity occurs."

3.5. Static allocation efficiency

The auction principle entails bid aggregation in a specific order prior to allocation. This ensures that the most suitable market participants are selected at that point of time: the bids of producers are sorted in an increasing order, i.e. the producers with the lowest willingness to sell are allocated first, and the bids of consumers are sorted in a decreasing order, i.e. the consumers with the highest willingness to pay are allocated first (Ehrenmann et al., 2019).

Continuous trading does not aggregate bids prior to allocation but follows the principle of "first-come, first-served" (IRENA, 2017). This feature cannot ensure that the most suitable market participants are matched because not all bid information are available at the time of allocation. For instance, this can lead to the situation that the consumer with the highest willingness to pay is not allocated to the producer with the lowest willingness to sell (instead to a producer with a sufficiently low willingness to sell) because of different bid submission times (Henriot, 2012).

An example of this reasoning is depicted in Fig. 1, inspired by Ehrenmann et al. (2019). The graphs illustrate the intraday market mechanism IDA on the left, and continuous trading on the right, where the ordinate denotes the price and the abscissa denotes the quantity.[10] The bids are given by the bars and they are the same in both graphs, i.e. six ask bids (colored grey) and seven offer bids (colored white).[11] In the case of IDAs, the bids of producers are sorted in an increasing order and bids of consumers in a decreasing order. The matching of these ordered bids is carried out as long as the willingness to pay is greater than the willingness to sell. In the example this holds for the first three bids leading to an allocated quantity in the auction of q_A and a uniform price

[6] Comprehensive overviews are given in Bellenbaum et al. (2014), Neuhoff, Richstein and May (2016b), Lanfranconi et al. (2019).

[7] Ehrenemann et al. (2019) refer to this difference as "competition on speed rather than on costs".

[8] Note that it is acknowledged that a high number of auctions may negatively affect the liquidity.

[9] Note that the information efficiency can be enlarged by the frequency of auctions, see for example IRENA (2017) or Lanfranconi et al. (2019).

[10] Note that for continuous trading the abscissa can also be interpreted as a timeline, representing the temporal order of bid submission.

[11] For the sake of readability we assume the same quantity of all bids (horizontal expansion).

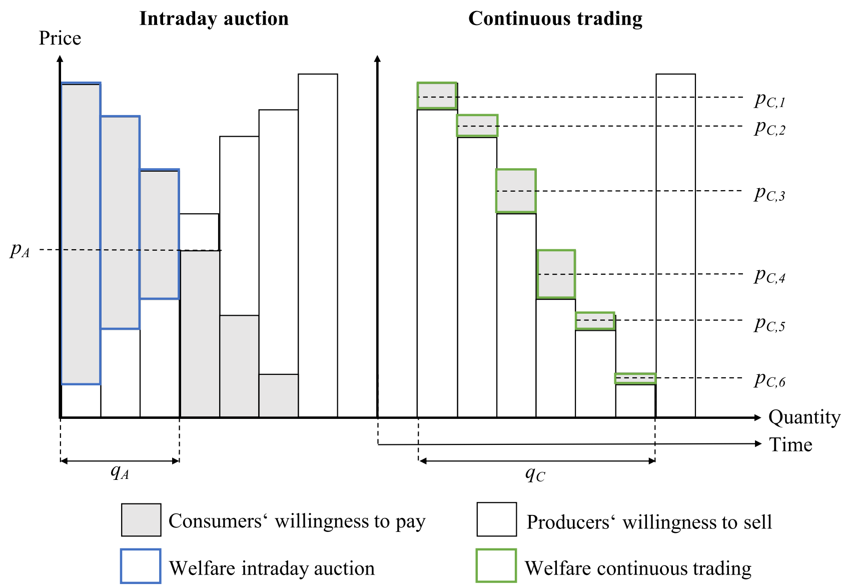

Fig. 1. Allocation example for IDAs and continuous trading.

of p_A (set by the highest rejected ask bid in our example). The resulting welfare is the sum of the blue framed areas: the consumers' surplus is the sum of the areas below p_A, and the producers' surplus is the sum of the areas above p_A.

When applying continuous trading, the bids of consumers and producers are not ordered prior to allocation but matched instantaneously in the order of bid submission as long as the willingness to pay is greater than the willingness to sell. In the example this leads to six allocations with a total quantity of q_C. Since pay-as-bid is applied, the resulting prices are different, ranging from $p_{C,1}$ to $p_{C,6}$.[12] The resulting welfare is the sum of the green framed areas, which, in this example, is smaller than the welfare of the IDA. Please note that this example depicts an extreme case because the trading principle of the XBID system is first-come first-served, i.e. the highest ask bid and the lowest offer bid get served first. Therefore, our example illustrates a rare case of bid submission timings in which the margins between ask and offer bids are small. This means that higher welfare solutions for continuous trading are possible.

3.6. Efficient usage of cross-zonal capacity

Lanfranconi et al. (2019) state: "the value of cross-zonal capacity can change from day-ahead to real-time according to updated information available to market participants and their willingness to pay." This means that the scarcity of cross-zonal capacity is time-dependent. Implicit continuous trading is unable to reflect these market dynamics efficiently because capacity is allocated for free as long as capacity is available (IRENA, 2017).[13] Consequently, the collection of congestion rents, i.e. the cross-zonal price difference times the allocated volumes, is not possible. In contrast, cross-zonal auctions correctly reflect network congestions because the bids are aggregated and cleared at one point in time. Lanfranconi, Lanza and Rossi (2019) conclude that an efficient allocation in a dynamic context can be guaranteed best by a high number of cross-zonal IDAs.

Table 1
Evaluation of market mechanisms (see also Bellenbaum et al., 2014).

	Continuous trading	Auction
Liquidity	–	++
Market power resilience	–	+
Information efficiency	++	–
Static allocation efficiency	–	+
Efficient usage of cross-zonal capacity	–	++

3.7. Summary: advantages and disadvantages

The following table summarizes the advantages and disadvantages associated with the two market mechanisms. The notation is inspired by Bellenbaum et al. (2014) and as follows: "+" (or "++") denotes that the criterion is (largely) met, "0" expresses neutrality and "-" (or "–") denotes a (largely) negative impact.

Table 1 illustrates the essential and unique advantage of continuous trading compared to the auction principle: information efficiency, i.e. enabling market parties to correct their position(s) in the market as soon as a change in their availability occurs. This possibility of instantaneous correction is associated with lower costs for the market parties and for the system, which in return is crucial for the efficiency of the market (e. g. IRENA, 2017; ACER/CEER, 2018). In conjunction with the generally viewed purpose of the intraday market until now (i.e. serving as an adjustment possibility), it is plausible why continuous trading is applied in most intraday markets in Europe at present.

4. Status quo in Europe: Empirical analysis

In this section we present the currently existing six IDAs in Europe. The structure of the analysis is presented in Table 2, while the market areas, their acronyms and the corresponding geographical scope are

[12] Pay-as-bid pricing incentivizes consumers and producers to shade their bids (i.e. consumers include a mark-down to their willingness to pay and producers include a mark-up their willingness to sell) in order to generate a profit in the case of allocation (e.g. Krishna, 2002). In the example this is considered by setting the resulting price in between the respective pair of willingness to pay and willingness to sell. Further note that the strictly declining order of prices is due to the example and cannot be assumed in general.

[13] Implicit means that transport capacity and electricity is traded in one step. This is commonly referred to as market coupling; for further information see APG (2020).

Table 2
Examined auction characteristics.

Number of auctions	It states how many auctions are carried out.
Tradable market period(s)	It states which market periods can be traded, e.g. the hours between 00:00 until 24:00 of delivery day D.
Market time unit	It states the granularity of the tradable market period(s), e.g. 15 min or 30 min.
Gate opening time	It states the time when the order book opens (also referred to as "opening of the order book").
Gate closure time	It states the time when the order book closes (also referred to as "closing of the order book").
Publication of results	It states when market results are published.
Admissible price interval	It states the price limits for bids.
Bid price granularity	It states the granularity of bid prices, e.g. 0.1 Euro/MWh (also referred to as "price increment").
Bid quantity granularity	It states the granularity of bid quantities, e.g. 0.1 MWh (also referred to as "volume increment").
Price rule	It states whether pay-as-bid pricing or uniform pricing is applied.

Table 3
Overview of currently existing European IDAs; sources: see Table A1 in the Annex.

Market area	Geographical scope
Germany (DE)	Germany, Luxembourg
Great Britain (GB)	England, Scotland, Wales
Iberian Peninsula (IP)	Portugal, Spain
Ireland (IR)	Northern Ireland, Irish Republic
Italy (IT)	Italy[a]
Switzerland (CH)	Switzerland

[a] Note that since June 2016 two of the seven IDAs are extended to the Italian-Slovenian border as part of a pilot project for implicit IDAs (e.g. Lanfranconi et al., 2019).

listed in Table 3.

Note that in the Annex (Table A1) the auction characteristics of the six market areas are presented in detail; we describe them briefly in the following.[14] Furthermore, Fig. 2 depicts information regarding the timings of the IDAs, i.e. for each auction the timespan between gate opening and closure as well as the tradable market period(s).[15]

The number of auctions ranges from one (DE) to seven (IT). The gate opening time varies between 45 days prior to delivery in DE to several hours prior in IT and IP. The gate closure time ranges from 03:00 pm day-ahead (e.g. DE or IT) to 05:00 pm on the delivery day in IR. The tradable market period(s) relate to the number of auctions as well as the gate closure times. For instance, in GB with two IDAs, there exist two different tradable market periods: for the first IDA with a gate closure at 06:30 pm day-ahead it includes the hours 00:00 until 24:00, whereas for the second IDA with a gate closure at 09:00 am it includes solely the hours 12:00 until 24:00. The publication of results is close to gate closure in all six market areas. The market time unit ranges from 60 min in IP, IT and CH, 30 min in GB and IR to 15 min in DE. Regarding the admissible price interval, DE has the highest limits with −3,000 Euro/MWh to +3,000 Euro/MWh, while in the market area IT it is +0 Euro/MWh to +3,000 Euro/MWh. The bid price granularity is either 0.1 Euro/MWh (DE, GB and CH) or 0.01 Euro/MWh (IP and IR), and the bid quantity granularity is 0.1 MWh in all five market areas for which information is available. Finally, all six market areas apply uniform pricing as price rule.

[14] Given the extent of the table, we decided to include Table A1 in the Annex and not in the main text. Note that for easier reading we use in the following description the acronyms of the market areas.

[15] Note that for the sake readability all gate opening times and gate closure times were harmonized to the Continental European Time.

5. The way towards European IDAs

There are two options for future European IDAs: cross-border IDAs and complementary regional IDAs. The legal stipulations for these two IDA types are presented in the following. Furthermore, we present IDAs that were recently approved under this new regulation.

5.1. Cross-border IDAs

On 24.01.2019 ACER published its decision "establishing a single methodology for pricing intraday cross-zonal capacity" required by Article 55 of the CACM Regulation (ACER, 2019a).[16] The methodology establishes a pricing mechanism for cross-border capacity in the intraday timeframe, including essential market design parameters for cross-border implicit European IDAs. These auction characteristics are presented in the following.

The methodology states that the pricing mechanism for cross-zonal capacity in the intraday timeframe shall be based on IDAs. These implicit auctions shall complement the already existing continuous SIDC. It is stated that cross-zonal capacity shall not be allocated to an IDA and continuous trading at the same time: trading for the continuous SIDC must be suspended temporarily. In line with the continuous SIDC, the TSOs responsible for a specific bidding zone border provide the available cross-zonal capacity as an input for the IDAs.

It is decided that one IDA shall be established on 03:00 pm D-1 (for all hours of the delivery day D), one IDA on 10:00 pm D-1 (for all hours of the delivery day D), and one IDA on 10:00 am on the delivery day (for the remaining hours of the delivery day D starting from 12:00 am). In addition, it is stipulated that the price rule shall be uniform pricing. Regarding price thresholds, the methodology refers to an ACER decision dated 14.11.2017 (ACER, 2017). In this decision and the corresponding methodology "harmonized maximum and minimum clearing prices for single intraday coupling" the maximum clearing price for SIDC is set to +9,999 EUR/MWh, and the minimum clearing price for SIDC is set to −9,999 EUR/MWh. The auction characteristics market time unit, gate opening time, publication of results, bid price granularity and bid quantity granularity are not specified.

Regarding implementation of cross-border IDAs, it is stated that the methodology shall "be implemented by amending and complementing the relevant requirements and methodologies related to the development of the SIDC" (ACER, 2019a). Thus, it does not state an exact date in the future. Table 4 summarizes these auction characteristics.

5.2. Complementary regional IDAs

Pursuant to Article 63 of the CACM Regulation, the relevant nominated electricity market operators (i.e. power exchanges such as EPEX or NordPool) and TSOs on bidding zone borders may jointly develop a methodology proposal for the design and implementation of complementary regional IDAs (henceforth referred to as "CRIDAs") and submit it to the concerned national regulatory authorities (henceforth referred to as "NRAs") for approval (European Commission, 2015). The proposals had to be submitted no later than 18 months after the entry into force of the CACM Regulation (dated 24.07.2015), i.e. until 24.01.2017. They may be implemented within or between bidding zones in addition to the continuous SIDC. As for cross-border IDAs, continuous trading within and between the relevant bidding zones may be stopped for a limited period of time in order to conduct the auctions. The respective NRAs can approve such a proposal for CRIDAs if certain conditions are met, amongst others: the CRIDAs may not have an adverse impact on the liquidity of the SIDC, they shall not introduce any undue discrimination

[16] Note that it is not in the scope of this paper to discuss the proposed solution ("hybrid model"). For further details please refer to Bellenbaum et al. (2014), ENTSO-E, 2017, IRENA, 2017 or Lanfranconi et al. (2019).

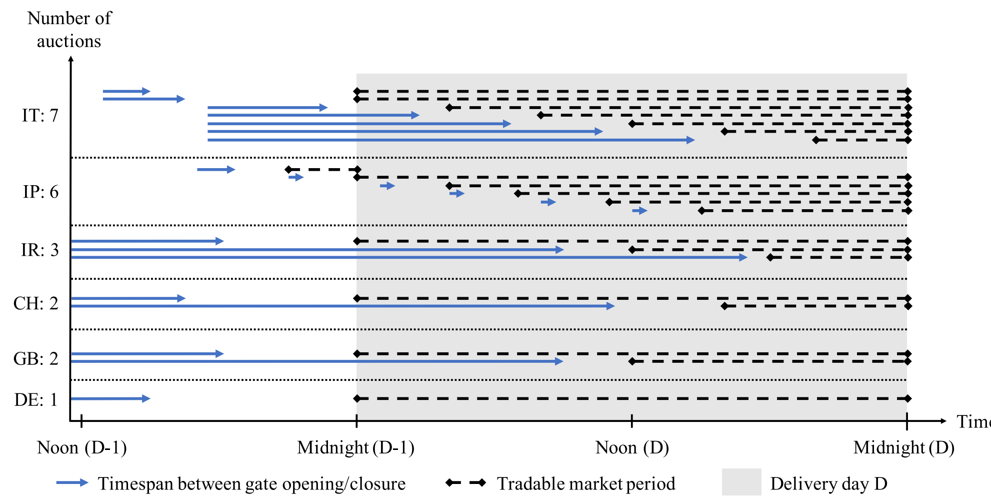

Fig. 2. Graphical presentation of IDA timings.

Table 4
Auction characteristics for cross-border IDAs according to ACER (2017, 2019a).

Number of auctions	3
Tradable market period(s)	00:00–24:00; 00:00–24:00; 12:00–24:00
Market time unit	*to be specified*
Gate opening time	*to be specified*
Gate closure time	03:00 pm (D-1); 10:00 pm (D-1); 10:00 am (D)
Publication of results	*to be specified*
Admissible price thresholds	−9,999 Euro/MWh/+9,999 Euro/MWh
Bid price granularity	*to be specified*
Bid quantity granularity	*to be specified*
Price rule	uniform pricing

between market participants from adjacent regions and the timetables for CRIDAs shall be consistent with SIDC to enable market participants to trade as close as possible to real-time. In order to ensure that these conditions are met also with proceedings SIDC developments, the respective NRAs shall review the compatibility of CRIDAs with the SIDC at least every two years after implementation.[17]

5.3. Approved CRIDAs

According to ACER's "monitoring report on the implementation of the CACM Regulation and the FCA Regulation", three proposals for CRIDAs were submitted and approved: one at the border of Portugal and Spain (approved on 13.04.2018), one by the capacity calculation region (henceforth referred to as "CCR") "Greece-Italy" (approved on 07.05.2019) and one by the CCR "Italy-North" (approved on 04.06.2019) (ACER, 2019b, 2020).[18] The Portuguese-Spanish CRIDAs are run already since the go-live of the XBID system in 2018 (CRIDA Approval ES-PT, 2018). We included their description in Section 4 by referring to this border as market area Iberian Peninsula. The characteristics of the CRIDAs for the CCR Greece-Italy and CCR Italy-North are identical, see Table 5 (CRIDA Approval GR-IT, 2019; CRIDA Approval IT-North, 2019).[19] In contrast to the implemented CRIDAs in the market area Iberian Peninsula, they are already aligned with the stipulations made by ACER regarding cross-border IDAs (see sections 4 and 5.1 for comparison). The implementation is foreseen as soon as both countries join the continuous SIDC end of 2020 (HUPX, 2019).

For Italy this would result in up to three different IDAs available for trading: the existing IDAs (Section 4), CRIDAs with the implementation of the continuous SIDC and cross-border IDAs with the methodology implementation for pricing intraday cross-zonal capacity. Regarding such a variety, ACER (2019b) suggests in its Monitoring Report of the CACM Regulation that CRIDAs should gradually be replaced by the methodology for pricing intraday cross-zonal capacity: "the need for additional CRIDAs might not be substantiated anymore". On the contrary, it could be beneficial if the concepts of CRIDAs and cross-border IDAs "are merged into a single methodology for pricing of intraday cross-zonal capacity, which should ideally be harmonized across the EU"

Table 5
Auction characteristics of CRIDAs in the CCR Greece-Italy and CCR Italy-North (CRIDA Approval GR-IT, 2019; CRIDA Approval IT-North, 2019).

	CCR Greece-Italy and CCR Italy-North
Number of auctions	3
Tradable market period(s)	00:00–24:00; 00:00–24:00; 12:00–24:00
Market time unit	60 min.
Gate opening time	01:00 pm (D-1); 03:30 pm (D-1); 10:30 pm (D-1)
Gate closure time	03:00 pm (D-1); 10:00 pm (D-1); 10:00 am (D)
Publication of results	03:30 pm (D-1); 10:30 pm (D-1); 10:30 am (D)
Admissible price thresholds	−9,999 Euro/MWh/+9,999 Euro/MWh
Bid price granularity	0.1 Euro/MWh
Bid quantity granularity	0.1 MWh
Price rule	uniform pricing

[17] Note that the methodology for pricing intraday cross-zonal capacity (as presented in Section 5.1) does not apply for CRIDAs (European Commission, 2015).

[18] The rationales behind the CRIDA implementations at these borders are described in the respective proposals and approvals (CRIDA Approval ES-PT, 2018; CRIDA Approval GR-IT, 2019; CRIDA Approval IT-North, 2019).

[19] The reason for this is that initially there was only one proposal for both CCRs submitted to the relevant NRAs. In the course of the approval process, they had to be split up and submitted separately (ACER, 2020).

to avoid "the risk of too fragmented intraday markets in terms of timings, design and geography" (ACER, 2019b). In fact, the CRIDA methodologies of CCR Greece-Italy and Italy-North exclude the possibility of having three different types of IDAs at the same time: as soon as the methodology for pricing intraday cross-zonal capacity is implemented, the CRIDAs shall be replaced by cross-border IDAs (CRIDA Approval GR-IT, 2019; CRIDA Approval IT-North, 2019). Against the background that the CRIDAs of the CCRs Greece-Italy and Italy-North are already in line with the stipulations for cross-border IDAs, no major problems in the transition should arise.

6. Conclusions and policy implications

In the last section we summarize the results, discuss policy implications and point to future fields of research.

6.1. Summary of results

This paper examined the status quo of intraday auctions in Europe and presented an outlook for future European intraday auctions. This paper analyzed existing national intraday auctions, the future pan-European cross-border intraday auctions and planned (or partly implemented) complementary regional intraday auctions. We compared the two market mechanisms continuous trading and auction and discussed major advantages and disadvantages. We presented the existing six European intraday auctions and compared crucial auction characteristics. A wide variety in auction designs was found. We outlined two recent decisions by the Agency for the Cooperation of Energy Regulators that provide the framework conditions for future cross-border intraday auctions as well as the relevant Article of the Guideline on Capacity Allocation and Congestion Management for the implementation complementary regional intraday auctions. Furthermore, it was discussed that the complementary regional intraday auctions of the border Portugal-Spain and the capacity calculations regions Greece-Italy and Italy-North are already approved.

6.2. Policy implications

It remains open whether the five intraday auctions described in Section 4 and the complementary regional intraday auctions in the market area Iberian Peninsula persist. For the former, no proposals to implement them as complementary regional intraday auctions were submitted to the relevant national regulatory authorities. Furthermore, and this also holds for the latter, several auction characteristics are not in accordance with the Agency for the Cooperation of Energy Regulators' legal stipulations for cross-border intraday auctions. Most notably, this includes the number of auctions and the associated tradable market period(s), gate opening and gate closure times (see Fig. 2).

Assuming that the existing intraday auctions persist, the smallest adjustment effort should arise for the market area Ireland with three intraday auctions because mostly timings need to be changed. For the other existing intraday auctions, the implementation of cross-border intraday auctions will require adaptations. In particular, it poses the question if and how to harmonize these auctions with respect to economical (e.g. market fragmentation), legal (e.g. future harmonized European legislation as indicated by the Agency for the Cooperation of Energy Regulators) and political (e.g. Great Britain leaving the European Union) developments. For the market area Germany, for instance, it needs to be clarified whether the current national intraday auctions is still needed when three cross-border intraday auctions are introduced, while for the market area Iberian Peninsula a decision is required whether only the three cross-border intraday auctions are sufficient or if the sum of six intraday auctions should persist in the future (i.e. retaining three complementary regional intraday auctions).

6.3. Directions for further research

Future research should investigate the co-existence of multiple intraday auction concepts (on a per country basis, bidding-zone border basis or capacity calculation region basis) more thoroughly. Our paper only set the starting point for such an examination by illustrating the status quo and the legal stipulations. The focus could be placed on effects regarding market liquidity (e.g. in continuous intraday trading, intraday auctions, complementary regional intraday auctions, cross-border intraday auctions and balancing energy market), bidding strategies across different trading options and implications regarding the remaining time for continuous trading.

CRediT authorship contribution statement

Fabian Ocker: Conceptualization, Methodology, Investigation, Resources, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration. **Vincent Jaenisch:** Methodology, Investigation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Annex.

Table A.1
Comparison of auction characteristics in the six market areas; sources: Germany: EPEX SPOT SE (2019, 2020b); Great Britain: SEMOpx (2017, 2020); Iberian Peninsula: Chaves-Ávila and Fernandes (2015) and OMIE (2018); Ireland: SEMOpx (2017, 2020); Italy: Oggioni and Lanfranconi (2015) and GME (2003, 2017, 2020); Switzerland: EPEX SPOT SE (2019).

	Germany	Great Britain	Iberian Peninsula	Ireland	Italy	Switzerland
Number of auctions	1	2	6	3	7	2
Tradable market period(s)	00:00–24:00	00:00–24:00; 12:00–24:00	21:00 (D-1) - 24:00 (D); 00:00–24:00; 04:00–24:00; 07:00–24:00; 11:00–24:00; 15:00–24:00	00:00–24:00; 12:00–24:00; 18:00–24:00	00:00–24:00; 00:00–24:00; 04:00–24:00; 08:00–24:00; 12:00–24:00; 16:00–24:00; 20:00–24:00	00:00–24:00; 16:00–24:00

(*continued on next page*)

Table A.1 (*continued*)

	Germany	Great Britain	Iberian Peninsula	Ireland	Italy	Switzerland
Number of auctions	1	2	6	3	7	2
Market time unit	15 min.	30 min.	60 min.	30 min.	60 min.	60 min.
Gate opening time	45 days prior to delivery	14 days prior to delivery	05:00 pm (D-1); 09:00 pm (D-1); 01:00 am (D); 04:00 am (D); 08:00 am (D); 12:00 am (D)	19 days prior to delivery	00:55 pm (D-1); 00:55 pm (D-1); 05:30 pm (D-1); 05:30 pm (D-1); 05:30 pm (D-1); 05:30 pm (D-1); 05:30 pm (D-1)	14 days prior to delivery
Gate closure time	03:00 pm (D-1)	06:30 pm (D-1); 09:00 am (D)	06:40 pm (D-1); 09:40 pm (D-1); 01:40 am (D); 04:40 am (D); 08:40 am (D); 12:40 am (D)	06:30 pm (D-1); 09:00 am (D); 05:00 pm (D)	03:00 pm (D-1); 04:30 pm (D-1); 11:45 pm (D-1) 03:45 am (D); 07:45 am (D); 11:15 am (D); 03:45 pm (D)	04:30 pm (D-1); 11:15 am (D)
Publication of results	03:10 pm (D-1)	09:10 pm (D-1); 09:40 am (D)	06:45 pm (D-1); 09:45 pm (D-1); 01:45 am (D); 04:45 am (D); 08:45 am (D); 12:45 am (D)	07:10 pm (D-1); 09:40 am (D); 03:15 pm (D)	03:30 pm (D-1); 05:00 pm (D-1); 00:15 am (D); 04:15 am (D); 08:15 am (D); 11:45 am (D); 04:15 pm (D)	04:45 pm (D-1); 11:30 am (D)
Admissible price interval	−3,000 Euro/MWh/ +3,000 Euro/MWh	−150 Euro/MWh/ +1,500 Euro/MWh	n/a	−150 Euro/MWh/ +1,500 Euro/MWh	+0 Euro/MWh/ +3,000 Euro/MWh	−500 Euro/MWh/ +3,000 Euro/MWh
Bid price granularity	0.1 Euro/MWh	0.1 Euro/MWh	0.01 Euro/MWh	0.01 Euro/MWh	n/a	0.1 Euro/MWh
Bid quantity granularity	0.1 MWh	0.1 MWh	0.1 MWh	0.1 MWh	n/a	0.1 MWh
Price rule	uniform pricing					

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